DECK-LOG

The boat’s records: deck, weather, engine and provisions

Version 1.1 — one log, append-only, UTC, offline. The same file whether you keep it at a prompt or on a tablet.

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1. What this is, and why it is built this way

DECK-LOG keeps the records a boat has to keep: where you were and what the weather was doing, what the engine has had done to it, and what is in the lockers. It prompts every three hours, because that is the interval a weather observation is worth having at.

Four constraints shaped it.

1. The log is evidence. Consequence: it is **append-only**. Nothing is ever edited and nothing is ever deleted. A correction is a new record that points at the old one, and both stay visible for ever. That is what makes the log worth anything in an inquiry.

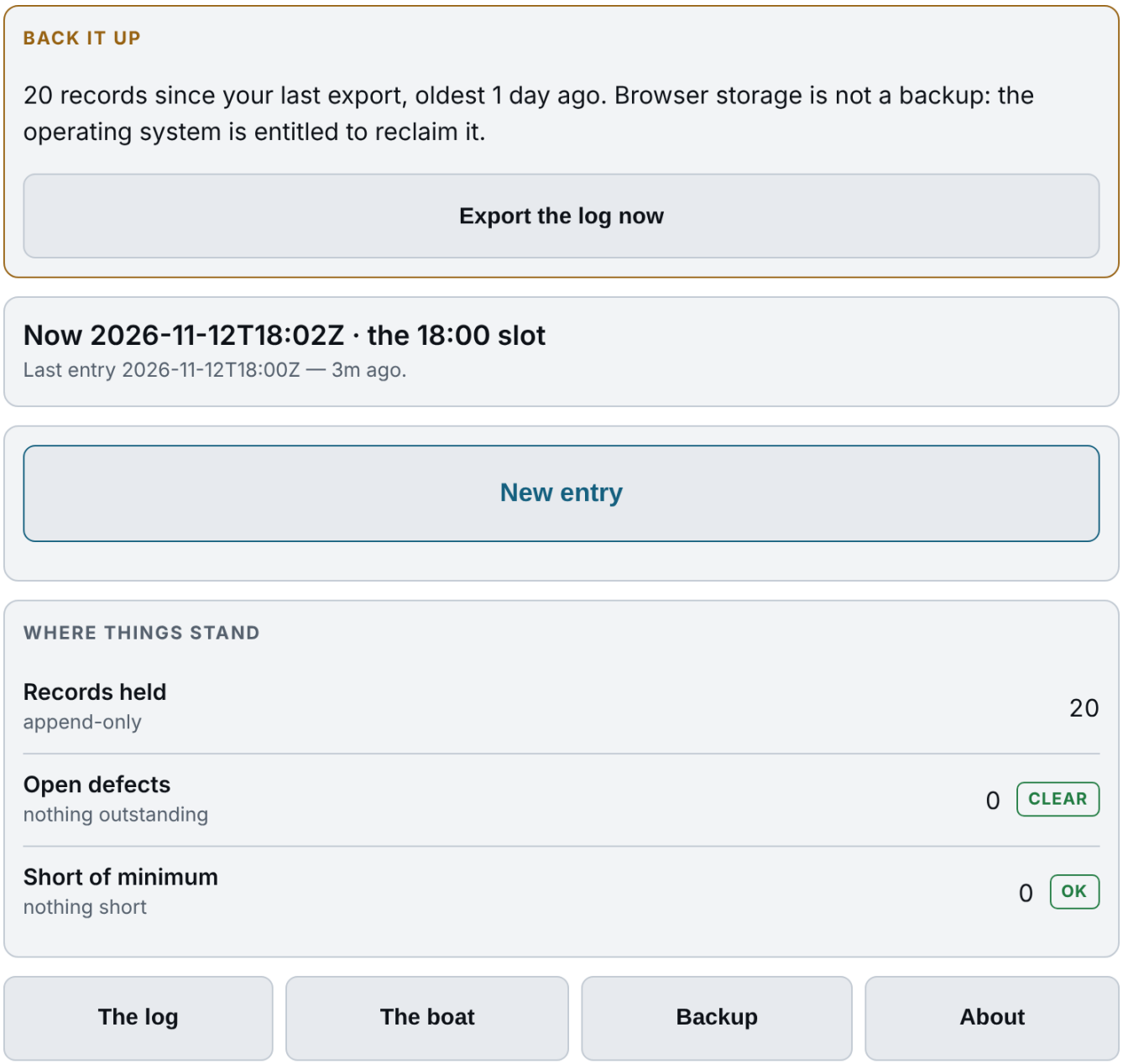
2. Two builds, one log. Consequence: the POSIX `awk` build and the browser build read and write the **same file, byte for byte** — not two formats that look alike. You can keep the log on a tablet at sea and read it at a prompt afterwards.

3. An interrupted write must be visible. Consequence: every record ends with a completeness marker. See section 3; it is the one irreversible decision in the whole tool.

4. It must say what it does not know. Consequence: a missed three-hourly slot is shown as missed. Logging now records *this* slot, not the ones gone by, and the tool says so rather than quietly filling a gap.

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2. The three-hourly rhythm



The log records what happened on this boat. Nothing else may write to it — no drill, no scenario, no prediction merely looked up. It is evidence after an incident, read by people who were not there. build 7 — 11 Nov 2026

Figure 1 — the home screen, showing the slot you are in, when the last entry was, and what the records say about the boat.

1. **The slot, not the clock.** Observations belong to the standard three-hourly slots — 00, 03, 06, 09, 12, 15, 18, 21 UTC — because that is what makes two observations comparable, and comparability is the whole of weather reasoning.
2. **Position, course and speed carry forward.** They are *state*: the entry shows them as they were last logged and return accepts them. You type what changed.
3. **Wind is TRUE, not apparent.** Apparent wind changes on every tack, and backing-versus-veering is the most useful rule there is — it only works on the true wind.

2026-11-12T18:00Z — entry for the 18:00 slot

Times are UTC. The record is UTC whatever you display.

Deck | Weather | Engine | Stores

THE DECK LOG

Position, course and speed are state: they are shown as they were last logged and return accepts them. Wind is TRUE, not apparent — apparent changes on every tack, and backing-versus-veering is the most useful rule there is.

Latitude e.g. 41 14.0N — carried forward

41 14.0N | - | /

Longitude e.g. 072 05.0W — carried forward

072 05.0W | - | /

Course steered, true degrees — carried forward

070 | - | /

Speed over ground knots — carried forward

6.5 | - | /

Wind direction, degrees TRUE 000–360 — last 176 (18:00Z)

| - | /

Wind speed knots — last 30 (18:00Z)

| - | /

Sea state

0 glassy | 1 rippled | 2 smooth, wavelets | 3 slight

4 moderate | 5 rough | 6 very rough | 7 high

8 very high | 9 phenomenal

- | /

Anything worth recording

Decline this section | Log this section

Figure 2 — an entry. Four sections: deck, weather, engine, stores.

What goes in each section

Deck	position, course steered, speed over ground, true wind direction and speed, sea state (Douglas), a note
Weather	pressure and its *characteristic*, the 3-hour change, air and dew point, sea temperature, total cloud, low / middle / high cloud, visibility, swell direction, period and height
Engine	which machine, running hours, work done, the part used and how many, fuel remaining, oil state, notes
Stores	water and gas remaining, stores used and how many, what is short

Why the weather section is longer than a deck log's usually is. The pressure *characteristic* (the shape of the last three hours, not the amount), the cloud types by level, and the swell period are exactly what the reasoning in `weather` needs and exactly what an ordinary deck log leaves out. The swell period is the one to log carefully: a long-period swell from a new direction is a system out there, because swell outruns the storm that made it.

Engine entries carry hours, not just dates. Maintenance runs on running hours, so the schedule is computable from the log with nothing else. The intervals come from your own engine manual, never from this tool.

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3. The record format, and the completeness marker

One record, one line:

```
YYYY-MM-DDThh:mmZ|type|key=value|key=value|...|~
```

The trailing `|~` is a completeness marker, and it is the reason the format survives an interrupted write. Kill the app mid-write and the half-line left behind is *structurally perfect*: right timestamp, right type, every field a well-formed `key=value`. Nothing distinguishes it from a finished record, so a reader accepts it — and you have silently lost whatever never reached storage. An engine entry with the part but not the hours, and no sign anything is missing. In an append-only log that is permanent, and it is exactly the failure that would matter in an inquiry.

A truncation cannot forge the marker, because `|` is percent-encoded inside values and so `|~` appears nowhere else.

The types: `nav` deck, `wx` weather, `eng` engine, `pro` stores, `inv` stocktake, `cor` correction, `mrg` merge, `cel` celestial, `tid` tide, `con` contact, `chart` chart.

Every timestamp is UTC, in exactly that 17-character form, whatever the screen displays.

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4. Reading the log

THE LOG, MOST RECENT FIRST

20 records. Nothing here was ever edited or deleted; a correction is its own record.

2026-11-12T18:00Z weather
mslp=1005.6 ptend=6 pchg=-1.8 airt=10.7 dewp=10.5 seat=12 cloud=8 cl=7 cm=2 ch=0 vis=4 swdir=228 swper=9 swht=3.2

2026-11-12T18:00Z deck
lat=41 14.0N lon=072 05.0W crs=070 sog=6.5 wdir=176 wspd=30 sea=5

2026-11-12T15:00Z weather
mslp=1007.4 ptend=6 pchg=-1.8 airt=10.8 dewp=10.5 seat=12 cloud=8 cl=7 cm=2 ch=0 vis=4 swdir=230 swper=9 swht=3

2026-11-12T15:00Z deck
lat=41 14.0N lon=072 05.0W crs=070 sog=6.5 wdir=190 wspd=29 sea=5 note=Reefed main and staysail. Barometer still going.

2026-11-12T12:00Z weather
mslp=1009.2 ptend=6 pchg=-2.3 airt=10.8 dewp=10.3 seat=12.1 cloud=8 cl=7 cm=2 ch=0 vis=4 swdir=230 swper=9 swht=2.8

2026-11-12T12:00Z deck
lat=41 14.0N lon=072 05.0W crs=070 sog=6.5 wdir=190 wspd=27 sea=5

2026-11-12T09:00Z weather
mslp=1011.5 ptend=6 pchg=-2.5 airt=10.9 dewp=9.8 seat=12.1 cloud=8 cl=6 cm=2 ch=0 vis=5 swdir=235 swper=10 swht=2.5

2026-11-12T09:00Z deck
lat=41 14.0N lon=072 05.0W crs=070 sog=6.5 wdir=195 wspd=25 sea=4

2026-11-12T06:00Z weather
mslp=1014 ptend=6 pchg=-2.2 airt=11 dewp=9.1 seat=12.2 cloud=8 cl=5 cm=2 ch=7 vis=7 swdir=240 swper=10 swht=2.1

2026-11-12T06:00Z deck
lat=41 14.0N lon=072 05.0W crs=070 sog=6.5 wdir=200 wspd=22 sea=4

2026-11-12T03:00Z weather
mslp=1016.2 ptend=6 pchg=-1.9 airt=11.2 dewp=8.4 seat=12.2 cloud=7 cl=5 cm=1 ch=7 vis=7 swdir=245 swper=11 swht=1.8

2026-11-12T03:00Z deck
lat=41 14.0N lon=072 05.0W crs=070 sog=6.5 wdir=210 wspd=19 sea=4

2026-11-12T00:00Z weather
mslp=1018.1 ptend=7 pchg=-1.5 airt=11.4 dewp=7.6 seat=12.3 cloud=6 cl=5 cm=0 ch=6 vis=8 swdir=250 swper=11 swht=1.5

2026-11-12T00:00Z deck
lat=41 14.0N lon=072 05.0W crs=070 sog=6.5 wdir=220 wspd=17 sea=3

2026-11-11T21:00Z weather
mslp=1019.6 ptend=7 pchg=-1.2 airt=11.5 dewp=6.9 seat=12.3 cloud=4 cl=1 cm=0 ch=4 vis=8 swdir=250 swper=11 swht=1

Figure 3 — the log, most recent first, as records rather than as prose.

The log view shows the records themselves. That is deliberate: the record is what is kept, so the record is what you check. Nothing in this view can edit anything.

Correcting an entry. `deck-log correct` (or the same from the menu) writes a **new** record of type `cor` that references the one it corrects. Both stay. A reader shows the correction against the original rather than replacing it, because which is which, and when each was written, is the point.

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5. The boat: registry, holdings, defects, shopping

OPEN DEFECTS

A finding is not an event that is over when it is written. It stays open until a job records that it closes it.

Nothing outstanding.

WHAT IS IN THE LOCKER

Derived by replaying the log, never stored. The count cannot disagree with the log, because it *is* the log.

No part has moved yet. A stocktake or a job puts something here.

SHOPPING LIST

Everything the man behind the counter will ask for, on one screen, with no signal.

Nothing short of its minimum.

EQUIPMENT

No equipment yet. A boat has a main engine, a generator, a windlass, a watermaker — add them and each gets its own spares.

Add equipment

PARTS

No parts yet. Capture one at the moment of use: in a chandlery, "an impeller" is useless and the manufacturer’s number is everything.

Add a part

THE REGISTRY AS TEXT

Facts about the boat, not counts. Edit it here and it is saved as typed — the same escaping rule as the log.

Save the registry

Figure 4 — what the records add up to: equipment, what is aboard, what is outstanding, and what to buy next port.

Nothing here is a second database. **Every one of these is derived from the log**: the holdings are stores received minus stores used, the open defects are jobs raised and not yet closed, the shopping list is what has fallen below what you said you wanted to carry. Add a record and the views move; there is nothing to keep in step by hand.

command	
`deck-log inspect`	run the engine checklist
`deck-log job`	record work done, and the part it used
`deck-log open`	what is outstanding
`deck-log spares`	what is aboard
`deck-log shopping`	the list for the next port
`deck-log equip` / `part`	the registry
`deck-log stocktake`	count the locker

6. Backup, export and merge

Browser storage is not a backup. The operating system is entitled to reclaim it, and it will not ask first. The home screen counts the records written since your last export and says so.

1. **Export** writes the whole log out as text — the same text the `awk` build reads.
2. **Import** brings a log back in, and **merges** rather than replaces: identical records appear once, and anything new is added in timestamp order. This is what lets the tablet and the laptop both keep the log on passage.
3. **The merge is previewed before it is applied**, with the count of records that are new, already held, in conflict, and any line it could not read at all.

A merge itself is recorded, as type `mrg`, so the log says when it absorbed another copy of itself.

7. Commands

`deck-log`	the menu
`deck-log entry`	a deck log entry — the three-hourly one
`deck-log wx`	a weather observation on its own
`deck-log log [n]`	the last n entries
`deck-log inspect`	run the engine checklist
`deck-log job`	record work done, and the part it used
`deck-log open`	what is outstanding
`deck-log spares`	what is aboard
`deck-log shopping`	the list for the next port
`deck-log equip` / `deck-log part`	the registry
`deck-log stocktake`	count the locker
`deck-log correct`	correct an earlier entry, properly
`deck-log day` / `night` / `plain`	colour mode
`deck-log about` / `version`	

The weather reasoning lives in its own tool. `deck-log` writes the observations; `weather` reads them. They share one log and one store, so nothing is copied between them.

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8. The display modes

The display: day, dark and night

The browser build has three display modes. They are not decoration: one of them exists to protect your night vision, and choosing the wrong one at the wrong moment costs you something real.

Mode	What it is for
NIGHT	Red on black, for a dark-adapted eye. The default.
DARK	Near-black with light text, for below decks, dusk, or indoors.
DAY	White with near-black text, for daylight and for printing.

Why night is the default

A tool you open on deck in the dark must not start bright. A bright panel costs a dark-adapted eye roughly **twenty minutes** to recover from, and twenty minutes is a long time to be unable to see a mark or a ship’s lights properly. So the first time you open any of these tools you get night, and from then on you get whatever you last chose.

Three buttons, not a cycle

The three modes are three separate buttons, and the one you are in is shown as selected. That is deliberate.

1. **A cycle can take you somewhere you did not intend.** One stray press with a cycling control can land you on DAY at three in the morning.
2. **A cycle cannot tell you where you are**, only what the next press would do. Three buttons show the current mode at a glance without changing it.
3. **Going darker is always one tap.** DARK and NIGHT are immediate from anywhere, because going darker never costs you anything.
4. **Going from NIGHT to DAY takes a deliberate half-second press.** Hold the DAY button and it fills as it goes; let go early and nothing happens. Nothing bright appears on the screen until the change actually completes. That one transition is guarded because that one transition is the only one that costs you your dark adaptation.

The setting follows you

The mode is remembered for all of the tools together, and for the course pages that embed them. Set night in one and you are in night in the next. It is remembered in the browser, so it survives closing the app.

It is right before the page draws

The mode is set in the first instructions the page runs, before any of it is painted. This matters on a first visit over a poor connection: set it later and the page would appear in one palette and then jump to another, which at night is the difference between keeping your night vision and losing twenty minutes of it. If nothing has been stored yet, or whatever was stored cannot be read, the page opens in **night**.

The controls follow the palette too

Sliders, scroll bars and the other controls the browser draws are told which world they are in, so they are drawn in the mode you chose rather than the browser’s own light default. Without that, a night screen can carry a bright white slider across the middle of it.

If you use High Contrast Mode

On Windows, High Contrast replaces every colour on the page with the ones your system theme specifies. That applies here too: the tools do not override it, because it is your setting and you chose it.

Nothing is lost by it. Where a colour carries meaning — the navigation lamps in colregs — the letter beside it says the same thing, so a lamp is still identifiable when the hue has gone.

One thing to know if you need both. High Contrast wins over night mode. If you need night vision on deck *and* high contrast to read, choose a dark high-contrast theme in Windows; the tools will follow it.

What the display can and cannot do in sunlight

Worth knowing before you blame the app on a bright day.

In sunlight, light reflecting off the glass lands on the dark parts of the screen and the light parts alike. That lifts everything together and flattens the difference between them, so the contrast you actually see collapses no matter what colours the page uses:

Condition	Body text, as seen
Indoors	about 10:1
Overcast on deck	about 6:1
Bright	about 3.5:1
Full sun on the glass	about 2.3:1

At the extreme, every colour scheme converges. Pure black on pure white measures the same 2.3:1 in full sun as the day mode does, because the glare swamps the difference. A different palette is not the answer, and nor is a different shade of white: across every plausible choice of background the contrast changes by under 4%, which is below what an eye can see.

What does work, in order:

1. **Turn the screen brightness up.** This is the one large lever, and it is the device’s, not the app’s.
2. **Shade the screen** with a hand, your body or the sprayhood. Removing the reflection is worth more than any change the app can make.
3. **Use DAY mode**, which at least puts the maximum luminance the panel can give behind the darkest text it can draw.

A note for printing

Documents and their figures are made in **day** mode. A screen captured in dark or night mode prints as a solid block of toner and loses the fine type in a working, so the printed pages show the day display even if that is not the one you use at sea.

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9. Offline, and putting it on the home screen

1. Open the browser build once with a signal.
2. **Share → Add to Home Screen.**
3. It then runs with the antenna down, keeps the log on the device, and needs no network at any point.

The log lives in the browser’s own database on that device. It is not sent anywhere, by this tool or by anything in it.

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10. Cautions

1. **Export regularly, and keep the export somewhere else.** Browser storage can be reclaimed by the operating system; a log you cannot produce is not a log.

2. **The log is not edited.** If something is wrong, correct it with a `cor` record. Editing the exported text by hand breaks the one property the format exists to give you.
3. **Maintenance intervals are yours**, from your engine's own manual. This tool computes against what you tell it, and knows nothing about your engine.
4. **A missed slot stays missed.** Logging later records the slot you are in. Do not back-date an observation you did not take.